DEPI PHASE 5 – GRADUATION PROJECT

Partner: CLS Learning Solutions

NabdMasr

منصة نبض مصر للبيانات المفتوحة

A Unified Open Data Platform for Egyptian Public Healthcare & Demographic Indicators

Project Name:	NabdMasr (نبض مصر)
Track:	Data Engineering & Business Intelligence
Team Leader:	Ahmed Daoud
Team Members:	Ahmed Hassan, Abdelrahman Nader, Ahmed Mohammed, Omar Ebied
GitHub Repository:	https://github.com/ahmeddaoud07/nabd-masr-data-platform
Submission Date:	October 2026

Ministry of Communications and Information Technology (MCIT)

Digital Egypt Pioneers Initiative (DEPI)

1. Project Planning & Management

1.1 Project Idea & Overview

Project Concept: *NabdMasr (نبض مصر)* is an open data platform aimed at gathering, cleaning, and unifying scattered Egyptian public data—with an initial focus on the healthcare and public health sector.

In Egypt, public data is often distributed across various official portals, statistical bulletins, and international repositories in multiple disparate formats (such as Excel spreadsheets, CSVs, and published PDF reports). In addition, differences in Arabic spelling (such as naming conventions for governorates) and varying levels of data quality make it difficult for citizens, researchers, and decision-makers to get a clear, consolidated view.

Proposed Solution: The project builds an automated data pipeline that:

- Extracts public healthcare data from authoritative sources (such as CAPMAS, data.gov.eg, WHO, and the World Bank).
- Cleans and standardizes the data (handling missing values, fixing character encodings, and unifying governorate names).
- Stores the processed records in a centralized data warehouse modeled for analytical reporting.
- Presents the findings through intuitive dashboards for citizens and decision-makers, with optional API access for developers.

1.2 Project Objectives

- **Centralize Fragmented Data:** Aggregate key healthcare and demographic indicators into a single accessible repository.
- **Improve Data Quality:** Automate data cleansing to ensure consistency across reporting years and geographical regions.
- **Enable Clear Visualization:** Deliver interactive dashboards that allow users to compare indicators across governorates.
- **Ensure Reproducibility:** Build the data pipeline using standard tools to enable scheduled updates with minimal manual effort.

1.3 Project Scope

In-Scope	Out-of-Scope (Future Enhancements)
• Public healthcare infrastructure metrics (e.g., hospitals, beds, units). • Demographic and health indicators across Egyptian governorates. • Automated extraction scripts for public tabular and report data. • Data cleansing, normalization, and relational modeling. • Interactive visualization dashboards (Power BI / Superset / Streamlit). • Data quality validation checks and documentation.	• Confidential or individual patient medical records (EMR/EHR). • Real-time hospital sensor or ICU telemetry streaming. • Private hospital financial accounting systems. • Direct integration with national civil registry databases.

1.4 Execution Schedule (6-Week Roadmap)

Timeframe	Phase	Key Objectives & Deliverables
Week 1	Planning & Discovery	Finalize scope, explore public data sources, set up repo, and define architecture.
Week 2 – 3	Ingestion & Cleansing	Build data extraction scripts, clean Arabic text, and unify governorate names.
Week 4	Data Modeling & Storage	Design warehouse tables (facts & dimensions) and implement transformation logic.
Week 5	Orchestration & Quality	Automate workflow execution and add automated data quality tests.
Week 6	Serving & Final Review	Build interactive dashboards, test endpoints, finalize documentation, and prepare demo.

1.5 Team Roles & Responsibilities

Team Member	Primary Role	Core Responsibilities
Ahmed Daoud	Team Leader / Data Modeling	Project coordination, architecture overview, dimensional modeling, and milestone tracking.
Ahmed Hassan	Data Pipeline & Ingestion	Building data connectors, scrapers, and raw storage integration.
Abdelrahman Nader	Data Cleansing & Transformation	Arabic text normalization, schema mapping, and cleaning routines.
Ahmed Mohammed	Warehouse & Database Management	Database setup, relational tables, and query optimization.
Omar Ebied	BI Visualization & Quality Assurance	Interactive dashboard design, visual analytics, and pipeline quality testing.

1.6 Risk Management & Mitigation

Identified Risk	Impact	Mitigation Strategy
Changes in source portal structure: Websites may update layouts or change links.	Moderate	Keep raw downloaded snapshots backed up; design modular scrapers that can be easily updated.
Arabic text & encoding issues: Differences in encoding or naming across entities.	Moderate	Use standardized normalization rules and a canonical list of Egyptian governorate names.
Missing data for some regions: Certain years or governorates may have incomplete stats.	Low	Clearly flag missing values in the visualization layer without altering original figures.

1.7 Project Success Criteria

- Successfully consolidating at least 3 distinct public data sources into a unified schema.
- 100% of recognized governorate entries mapped to standard canonical names.
- Clean, interactive dashboards presenting key metrics in a simple and intuitive format.
- A documented, reproducible workflow that can be easily demonstrated and extended.

2. Literature Review

2.1 Background & Context

Open government data is an important foundation for public transparency, economic research, and informed decision-making. Worldwide, national health observatories compile fragmented public health indicators into central portals so that policymakers can monitor capacity and researchers can examine regional trends.

In Egypt, substantial statistical data is produced regularly by organizations such as the Central Agency for Public Mobilization and Statistics (CAPMAS) and the Ministry of Health. However, much of this information is shared through separate reports, yearbooks, or static web pages. Bringing these sources together into a single, structured platform helps unlock the true value of existing open data.

2.2 Review of Similar Solutions & Portals

1. Egyptian Open Data Portal (data.gov.eg)

Overview: The official national portal aimed at sharing open datasets from various government ministries.
Advantages: Official governmental platform covering multiple sectors.
Limitations: Datasets are often uploaded as separate, static files without unified schemas, automated cross-source joins, or interactive analysis tools.

2. CAPMAS Statistical Yearbooks & Bulletins

Overview: The official national statistical agency publishes annual bulletins on health services, hospitals, and population.
Advantages: Highly authoritative and detailed data covering all governorates.
Limitations: Data is primarily distributed in PDF format or complex tables, requiring manual compilation for multi-year trend analysis.

3. WHO Global Health Observatory (GHO)

Overview: The World Health Organization's central repository for global health metrics and country profiles.
Advantages: Internationally standardized metrics and programmatic API access.
Limitations: Focuses mainly on national-level aggregates for Egypt and lacks granular governorate-level breakdowns.

4. Our World in Data (Global Health)

Overview: An independent open-access platform known for clear visualizations and interactive charts on global development.
Advantages: Highly engaging visuals and transparent data curation.
Limitations: Global scope without localized Egyptian regional administration details.

2.3 Comparative Analysis

Feature	data.gov.eg	CAPMAS Reports	WHO GHO	NabdMasr (Our Project)
Governorate Breakdown	Partial	Yes (in PDFs)	National only	**Unified across 27 Governorates**
Automated Cleaning	No	No	Standardized	**Automated Normalization**
Unified Warehouse Schema	No	No	Relational	**Central Analytical Model**
Interactive Dashboard	No	No	Yes	**Simple, Focused Dashboards**

2.4 Differentiation & Project Value

- **Consolidation:** Merges multiple disparate public sources into one unified analytical view.
- **Standardization:** Resolves Arabic naming inconsistencies across governorates and reporting years.
- **Simplicity & Focus:** Focuses on delivering clear, actionable indicators rather than overwhelming raw tables.
- **Traceability:** Keeps track of original sources so users know where each metric originated.

2.5 Key References

1. Central Agency for Public Mobilization and Statistics (CAPMAS) – Annual Health Services Statistics Bulletins.
2. Egyptian Open Data Portal (data.gov.eg) – Ministry of Communications and Information Technology.
3. World Health Organization (WHO) – Global Health Observatory (GHO) data repository.
4. World Bank Open Data – World Development Indicators (Egypt Health & Demographics).
5. Kimball, R., & Ross, M. – The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling.

3. Requirements Gathering

3.1 Stakeholder Identification & Personas

Beneficiary Group	Their Needs	How the Project Solves It
Citizens & General Public	Clear, straightforward insight into public healthcare services across governorates.	Visual, interactive dashboards that are easy to explore without technical expertise.
Researchers & Students	Clean, multi-year statistical data ready for analysis.	Normalized datasets in accessible formats (CSV/database) with clear source attribution.
Healthcare Planners & Decision-Makers	High-level comparative view of capacity and resources across regions.	Summary KPI cards, governorate comparisons, and historical trend indicators.

3.2 Data Sources

- **CAPMAS:** Annual statistical bulletins covering hospitals, bed capacities, and health personnel.
- **Open Data Egypt (data.gov.eg):** Public directories of health units and facilities.
- **WHO & World Bank APIs:** Key national healthcare indicators and demographic trends for Egypt.

3.3 Functional Requirements

Area	Functional Requirement
Data Collection	The system shall ingest raw public data from designated files, APIs, or portals.
Data Cleansing	The system shall handle missing values, resolve Arabic text encoding, and remove duplicates.
Geographic Standardization	The system shall map governorate names to a canonical standard list to enable joins.
Data Storage	The system shall structure cleaned data into structured tables (facts & dimensions).
Data Visualization	The system shall present key indicators (e.g., beds, hospitals, trends) in interactive charts.
Quality Verification	The pipeline shall verify basic data integrity (no negative numbers, valid relationships).

3.4 Non-Functional Requirements

Category	Requirement Specification
Usability	Dashboards should be simple, intuitive, and easy to read for non-technical users.
Data Accuracy	Preserve original numbers during cleaning with transparent transformations.
Maintainability	Modular codebase so new sources or years can be added easily.
Security & Privacy	Rely solely on public, aggregated information; no personal identifiable data.
Reproducibility	Pipeline should run in a standard container or virtual environment with documented setup.

3.5 High-Level Architecture Overview

The overall architecture follows a clean, modular data flow:

Data Flow:
1. **Sources Layer:** Public portals & reports (CAPMAS, data.gov.eg, WHO, World Bank)
   ↓
2. **Ingestion & Cleansing:** Extraction scripts + Arabic text & governorate normalization
   ↓
3. **Data Warehouse:** Structured dimensional model (Fact tables & Dimension tables)
   ↓
4. **Serving & Visualization:** Interactive dashboards (Power BI / Superset) & optional API

Repository & Deliverable:
Official GitHub Repository: https://github.com/ahmeddaoud07/nabd-masr-data-platform
Submitted for DEPI Phase 5 Graduation Project.